

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 3

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August 7th, 2026



The Monty Hall Problem

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- What do you do? Does it make any difference?

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- If the money is in B , what is the probability that he opens box B ?
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- If the money is in A , what is the probability that he opens box B ?
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- If the money is in B , what is the probability that he opens box B ?
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- If the money is in A , what is the probability that he opens box B ?
 - $P(H_B|M_A) = 1/2$ (and similarly, $P(H_C|M_A) = 1/2$).
- If the money is in B , what is the probability that he opens box B ?
 - $P(H_B|M_B) = 0$ (and similarly, $P(H_C|M_B) = 1$).
- If the money is in C , what is the probability that he opens box B ?
 - $P(H_B|M_C) = 1$ (and similarly, $P(H_C|M_B) = 0$).

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- We know that $P(M_C) = 1/3$, and $P(H_B|M_C) = 1$.
- By the law of total probability,

$$\begin{aligned}P(H_B) &= P(H_B|M_A)P(M_A) + P(H_B|M_B)P(M_B) + P(H_B|M_C)P(M_C) \\&= \frac{1}{2} \times \frac{1}{3} + 0 \times \frac{1}{3} + 1 \times \frac{1}{3} \\&= \frac{1}{2}\end{aligned}$$

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- So, $P(M_C|H_B) = \frac{\frac{1}{3} \times 1}{\frac{1}{2}} = \frac{2}{3}$

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- So, we should switch!

Definition

Two events $A \subset \Omega$ and $B \subset \Omega$ are said to be independent if

$$P(A \cap B) = P(A)P(B)$$

Statistical Independence of Events

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- Thus, Independence implies that probability of occurrence event A does not change given event B has occurred.
- Events A and B are said to be **dependent** if they are not independent.

- Suppose we toss two fair dice. Let E_1 be the event that the sum of dice is six and F denote the event that the first die equals four. Are E_1 and F independent?

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- Suppose we toss two fair dice. Let E_1 be the event that the sum of dice is six and F denote the event that the first die equals four. Are E_1 and F independent?
- Let E_2 be the event that the sum of the dice equals seven and F denote the event that the first die equals four. Are E_2 and F independent?

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Proof:

- Independence of A and B^c :
 - We know that $A = (A \cap B) \cup (A \cap B^c)$.

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- Independence of A and B^c :
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 $P(A) = P(A \cap B^c) + P(A \cap B)$.

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- For other cases, do it yourself.

Mutual Independence

Question: Given three events A , B and C , s.t.

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- How about $P(A \cap B)$?
- $P(A \cap B) = P(\{\text{First roll is } 1 \text{ or } 3\}) = \frac{12}{36} = \frac{1}{3}$.
- But, $P(A) \times P(B) = \frac{1}{4}$. So, $P(A \cap B) \neq P(A)P(B)$.

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Actual Definition of Mutual Independence

The events $A_1, \dots, A_n$ are said to be independent if for every subset $\{i_1, i_2, \dots, i_k\}$ ($k \leq n$) of the set $\{1, \dots, n\}$, the events $A_{i_1}, \dots, A_{i_k}$ satisfy the following property.

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \dots P(A_{i_k})$$

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$$P(A_{i_1} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \dots P(A_{i_k})$$

- Thus, events $A_1, \dots, A_n$ are independent if the knowledge of occurrence of any of these events has no effect on the probability of any other event.

Mutual Independence of n Events

- To check the independence of n events $A_1, \dots, A_n$, we must check the following $2^n - n - 1$ relations:

$$\begin{cases} P(A_i \cap A_j) = P(A_i)P(A_j), & i \neq j; i, j = 1 \dots n \\ P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k), & i \neq j \neq k; i, j, k = 1 \dots n \\ \vdots \\ P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2) \dots P(A_n) \end{cases}$$

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- Does mutual independence implies pairwise independence? **yes!**
- Does the converse hold?

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- Thus, $A = \{HH, HT\}$, $B = \{HH, TH\}$ and $C = \{HH, TT\}$.

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- This brings us to conditional independence.